

Sai Manoj Varikuntla

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🔗 VARIKUNTLASAIMANOJ

EDUCATION

B.Tech Computer Science and Engineering - AIML

Malla Reddy University

Sept 2022 – June 2026

CGPA: 8.09/10

Intermediate Education

Narayana Junior College

2020 – 2022

Marks: 788/1000

Secondary Education

Gowtham Model School

2020

CGPA: 9.7/10

Experience

ICRISAT

AI/ML Intern

Feb 2026 – Present

Hyderabad, India

- Built a Retrieval-Augmented Generation (RAG) system using LangGraph, Qdrant, and locally hosted Llama models to enable natural language querying over a large scientific research corpus — achieving 80%+ response accuracy with a custom 6-layer evidence validation pipeline to minimize hallucination, with no dependency on external APIs.
- Developed an agro-meteorological intelligence platform for Telangana combining 30-year ERA5 reanalysis validation, GRU/LSTM deep learning forecasting, FAO-56 irrigation advisory logic, and an NVIDIA Earth-2 (Four-CastNet 3 + CorrDiff) real-time forecasting web app — resolving forecasts down to mandal level across all 33 Telangana districts, with 110 automated tests passing.

PUBLICATIONS

AI-Powered Shopping Assistants Using Sentiment Analysis

Apr 2025

Prof. Sabyasachi Chakraborty, V. Sai Manoj, Tejaswi Mahadev, U. Pranay Reddy, T. Vishal Prem, T. Vamshi Krishna

[10.56025/IJARESM.2025.1304252562](https://doi.org/10.56025/IJARESM.2025.1304252562) [🔗](#)

SKILLS

Programming Languages: Python, Java

Data Analytics & ML: Pandas, Numpy, Matplotlib, NLP

Frontend Development: HTML, CSS, Streamlit, React.js

Databases: MySQL

AI Frameworks & Concepts: RAG, LangChain, API Integration

Tools & Technologies: Git, GitHub, Postman, PowerBI, Google Cloud Platform (GCP)

PROJECTS

ShopScribe - AI-Powered Shopping Assistants Using Sentiment Analysis

[ShopScribe](#) [🔗](#)

- Developed an AI-powered Streamlit web app integrating Firecrawl API, Google Gemini, and VADER Sentiment to scrape and analyze product listings from Amazon, Flipkart, and Myntra.
- Delivered real-time sentiment insights, trend visualization, and aspect-based analysis for structured e-commerce intelligence.

Tejoraj

[Tejoraj](#) [🔗](#)

- Built a full-stack AI space science platform using React, FastAPI & LangGraph, integrating real-time NASA APIs (APOD, NEO, Image Library) and arXiv for multi-agent, source-cited responses with configurable reasoning modes
- Developed key platform modules including a research paper generator (PDF/DOC export), a Space Explorer dashboard with live NASA imagery & 3D solar system visualization, and Supabase-based authentication

CERTIFICATIONS & ACHIEVEMENTS

- Coursera Certification: Introduction to Python Programming, Introduction to Data Analytics
- Postman Certification: API Fundamentals Student Expert
- Secured 2nd Prize in National Level Hackathon Conducted By Malla Reddy University
- AgentX 24-Hour Innovation Hackathon Winner (Business Automation), Anurag University, Salesforce; designed and implemented an AI-driven customer onboarding & CRM workflow using Agentforce and Data Cloud.